

Next-Generation Convolutional Neural Networks for Automated Cattle Disease Diagnostics

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Abstract—Early detection of disease in cattle may prevent a farm from being completely devastated by two of the most severe illnesses: Lumpy Skin Disease (LSD) and Foot-and-Mouth Disease (FMD). A diagnostic tool based on the ConvNeXt-Tiny architecture was developed during this project. Using training methods such as Focal Loss and Cosine Annealing allowed for the development of an extremely sensitive model for detecting diseased animals. At test time, the final model achieved a test accuracy of 95.71%, and at peak validation, it reached an accuracy of 98.14%. The model successfully detected 98.23% of all FMD cases and 96.70% of all LSD cases.

Index Terms—Deep Learning, ConvNeXt, Precision Agriculture, Cattle Disease, Focal Loss, Veterinary AI

I. INTRODUCTION

A single ill cow can be responsible for a farm-wide disease outbreak, especially if the disease is as virulent and contagious as Foot-and-Mouth Disease (FMD) or Lumpy Skin Disease (LSD) [1]. As such, these diseases have the potential to ruin livelihoods through the destruction of livestock. Livestock are ruined when the animals stop producing milk, lose weight due to a lack of appetite caused by illness, and sometimes require mass slaughter in order to prevent the spread of the disease. Traditional methods of detecting this type of illness include visually checking the animal (e.g., looking for drool, skin bumps, etc.). However, due to the size of many farms, it is unrealistic to expect a farmer to closely inspect all animals each day.

Computer vision has been proposed as a method for detecting sick cows. Early versions of computer vision-based cow detection systems were typically based on one of two types of networks: ResNet [2] or standard Convolutional Neural Networks (CNNs) [3], both of which had limited success processing images of cows taken from blurry photos or where the background was muddy. The version of ConvNeXt [4] we developed utilizes a modern network architecture that combines traditional CNN features with those developed using Vision Transformers [5], making ConvNeXt particularly well-suited for pulling information from messy backgrounds. In addition to the ConvNeXt model architecture, we also utilized Focal Loss [6] during training to prioritize identifying cows that are actually sick (thereby minimizing the number of false negatives), and we employed Cosine Annealing [7] to regulate the rate at which the model learns.

II. RELATED WORK

Precision livestock farming has seen rapid growth. Initial work centered on facial recognition to track eating and drinking behaviors [8]. Object detection models like YOLO [9] are currently being used to draw bounding boxes around lesions; however, they can have difficulty understanding overall image context. Studies using image classification techniques and ResNet-50 have achieved approximately 92% accuracy [10].

However, real-world veterinary datasets experience severe class imbalance (thousands of healthy images and very few diseased images). As a result, networks often learn to simply guess that an image is healthy most of the time. This project applies Focal Loss [6]—originally designed for street scenes with many background objects and only a handful of rare target objects—combined with ConvNeXt [4] to address the class imbalance issue and push accuracy higher.

III. METHODOLOGY

A. Data and Preparation

There are three types/categories to consider with this data set: Foot-and-Mouth disease, Lumpy skin disease and Healthy cattle. A total of 3,244 images were used in this study that were obtained from kaggle [11] and can be categorized into three different class labels; FMD (746 images), LSD (1,207 images) and Healthy (1,291 images).

To accommodate different types of farm lighting and camera angles, data augmentation was performed on the training images (random flips, slight rotations, and random changes to brightness and contrast). All images were resized to an equal dimension and normalized to ensure the network could process them efficiently.

B. Network Architecture: ConvNeXt-Tiny

The ConvNeXt-Tiny [4] model contains approximately 27.8 million parameters in total. The first layer is referred to as the “Patchify Stem,” which cuts the image into 4×4 sections. After that, the data moves through four stages of convolutional layers.

As seen in Figure 4, each subsequent layer is able to find more complex patterns than the previous one (from basic edges to complex skin lesion textures). Since transfer learning was utilized, we initialized the network with weights pre-trained

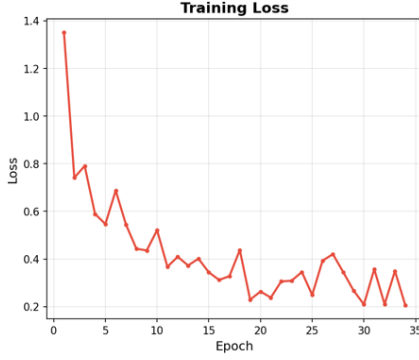


Fig. 1. Training and validation loss steadily decreasing over 34 epochs.

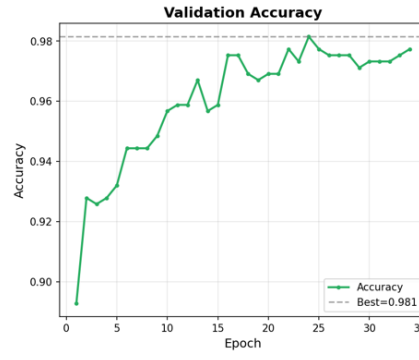


Fig. 2. Validation accuracy climbing and stabilizing near 98%.

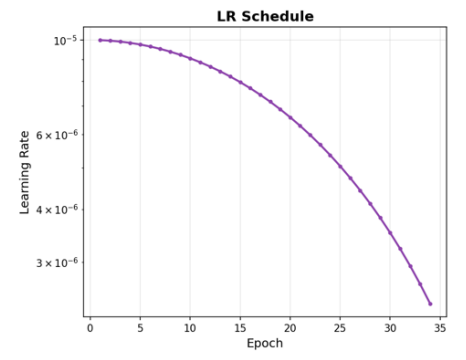


Fig. 3. Cosine annealing schedule smoothly reducing the learning rate.

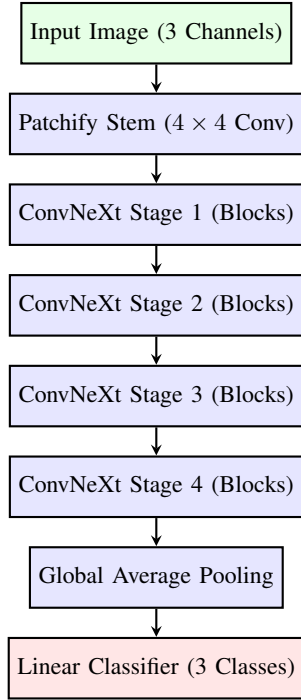


Fig. 4. The ConvNeXt-Tiny architecture used for disease classification.

on ImageNet. We froze 75% of the parameters and made all the remaining layers (approximately 7 million parameters) trainable so they could be fine-tuned specifically for cattle disease classification.

C. Mathematical Formulation: Loss and Learning Rate

The standard cross-entropy loss function is easily overwhelmed by the majority class (Healthy cows). In response to this issue, we employed Focal Loss [6]. A modulating factor is added to the standard cross entropy criterion in the mathematical definition of Focal Loss:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (1)$$

In the formula above, the model's estimated probability for the ground truth class, the weighting factor for class imbal-

ance, and the tunable focusing parameter are used. When the network can identify a healthy cow with ease, the modulating factor will go to zero, awarding the network little or nothing for this identification. However, when the network misses an easy-to-miss (subtle) FMD case, it will be heavily penalized and therefore forced to focus its deep attention on the difficult cases.

We also used a Cosine Annealing [7] learning rate schedule to regulate the speed of learning. At any given epoch, the learning rate is updated using the following equation:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos \left(\frac{T_{cur}}{T_{max}} \pi \right) \right) \quad (2)$$

Using the maximum and minimum learning rates, along with the current epoch and total number of epochs, training begins at high speed in order to learn large concepts as quickly as possible. After that, the curve declines smoothly so that the last few epochs take one small precise step at a time in order to fine-tune without losing what has been learned.

D. Experimental Setup

To ensure reproducibility, the specific training parameters utilized for the final model convergence are detailed in Table I. The model utilized the AdamW optimizer to handle weight decay effectively.

TABLE I
MODEL AND TRAINING HYPERPARAMETERS

Parameter	Value/Configuration
Base Architecture	ConvNeXt-Tiny
Pre-trained Weights	ImageNet-1K
Total Parameters	27.8 Million
Trainable Parameters	≈ 7.0 Million
Initial Learning Rate (η_{max})	1×10^{-4}
Total Epochs (T_{max})	34
Loss Function	Custom Focal Loss
Optimizer	AdamW

IV. RESULTS AND DISCUSSION

A. Evaluation Metrics

In order to accurately assess the performance of the model, we used precision, recall, and the F1-score. Depending solely on raw accuracy can be highly misleading in un-balanced veterinary data sets. The metrics are defined mathematically using True Positives (TP), False Positives (FP), and False Negatives (FN):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

The total test accuracy for this model was 95.71% and the Macro F1 score was 95.93%. During the experiment, it attained its highest peak validation accuracy of 98.14%.

B. Progression of Training

Loss was very high at Epoch 1 (1.35), and accuracy was approximately 89% at that time. Loss declined by nearly half (to 0.54) by Epoch 5, where validation accuracy first broke the 93% barrier. The Cosine Annealing [7] curve dictated how fast the learning rate decreased from initial values near 9.99×10^{-6} to slower rates over the epochs (see Figures 1, 2, and 3).

C. Performance Per Class

Table II and Figure 5 detail the performance across conditions. It is all about recall: the model captured nearly all of the FMD cattle (98.23%) and nearly all of the LSD cattle (96.70%). LSD had the least amount of precision (94.12%), meaning that on occasion, it labeled a healthy cow as sick.

TABLE II
DETAILED PERFORMANCE METRICS BY CLASS

Condition	Precision (%)	Recall (%)
Foot-and-Mouth (FMD)	96.52	98.23
Lumpy Skin Disease (LSD)	94.12	96.70
Healthy Cattle	96.81	93.33

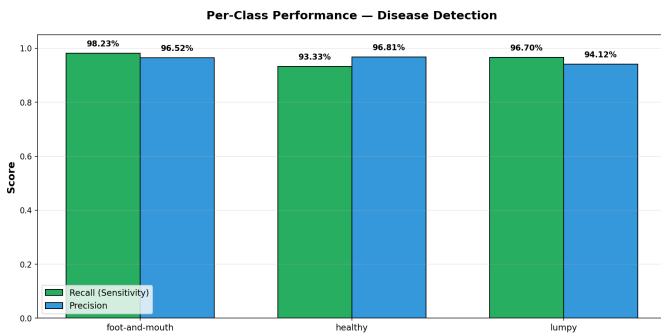


Fig. 5. Visual breakdown of Precision and Recall. Notice the incredibly high recall bars for both FMD and LSD.

D. Analysis of Error Types

Healthy cattle were most often incorrectly classified as LSD by this model (Figure 6). It was expected that the model would easily separate these classes because the raised skin lesions caused by LSD are visually distinct. However, on an actual farm, many cows are covered in dried mud, burrs, or ticks. These anomalies look like lesions to a camera and are thus misclassified by the highly cautious, Focal Loss-trained model as LSD.

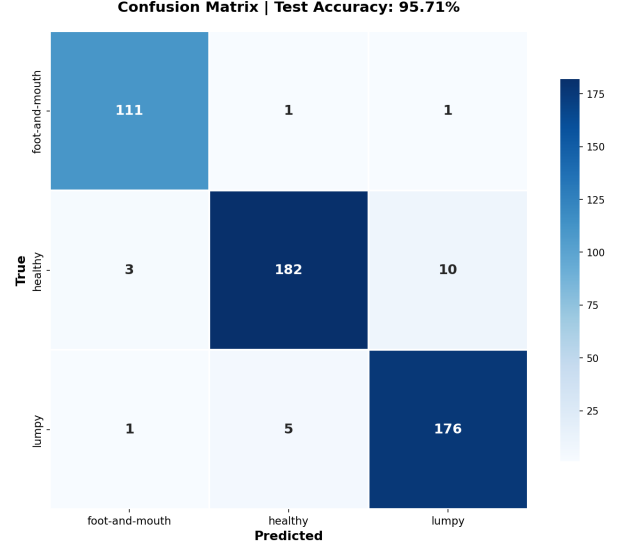


Fig. 6. The Confusion Matrix reveals the model's cautious nature. The most common error is misclassifying a healthy cow as having LSD.

Thus, the Healthy class experiences the lowest recall at approximately 93.33% (about 6-7 of every 100 healthy cows will be recommended for examination). A veterinarian can spend 5 minutes inspecting a muddy cow. That is vastly better than allowing a disease to spread due to a missed detection.

V. FARM DEPLOYMENT AND FUTURE WORK

Testing a model in a lab environment is merely a starting point; the true measure of effectiveness is how well the model functions when installed on a fence post at an active working dairy farm.

One major benefit of the relatively small number of parameters (27.8M) in our ConvNeXt-Tiny model [4], is that the model is large enough to be used with edge-based hardware (which includes ruggedized computers that may be mounted near farm based cameras), yet small enough to operate independently without needing a high-speed, continuous Internet connection. Therefore, all image processing occurs remotely as part of each video frame analysis. The remote location sends a brief notification to the farmer's mobile device in response to detecting a potential issue.

Our future work will involve developing a larger training set. Although we have successfully demonstrated detection of FMD and LSD using our model, there are numerous other diseases afflicting cattle including pinkeye and ringworm. In

addition, we intend to incorporate time-related information. One still image represents a moment in time. If you provide a short video clip to the model, it has the capability to analyze the cow's walk patterns, detect lethargic behavior or walking abnormally, which are common early indicators of disease prior to visual manifestation of lesions on the skin.

VI. CONCLUSION

AI could revolutionize animal husbandry. We made a robust tool by developing a model trained to avoid false positives using ConvNeXt [4], with careful use of Focal Loss [6]. The model had a high level of accuracy (95.71%) and very high levels of recall (over 98%), for detecting Foot and Mouth Disease. This means automated veterinary diagnostics are no longer only theoretically viable, they are also now highly accurate.

Although in some cases the model incorrectly identifies a dirty cow as sick, this cautionary approach can help protect against widespread disease on farms. As camera prices drop and the cost of computing decreases at the 'edge', models such as this will eventually leave laboratories and begin to monitor the herds.

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